Forecast-Informed Deep Learning for Electricity Price Prediction in the Australian NEM

Group 12
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Abstract

Spot electricity prices in Australia's National Electricity Market (NEM) are highly volatile, non-stationary, and prone to extreme price spikes. Using publicly available AEMO market data, we propose to implement and evaluate deep learning architectures for price forecasting across the five NEM regions, building from the recurrent and convolutional models covered in this course towards a graph-attention, multi-task architecture inspired by Li et al. (2025), benchmarked against the comparison of Gani et al. (2026).

1 Topic

We propose to implement a deep-learning based short-term electricity price forecasting system for the Australian NEM, which is composed of five interconnected regional markets (NSW, QLD, SA, TAS, VIC). We will reproduce a simplified version of the multi-task graph adaptive learning (MGAAL) model of Li et al. (2025), which frames price forecasting as a multi-task problem: a primary regression task predicting a future regional price, and an auxiliary classification task predicting anomalous price-spike events, with the two tasks sharing a graph-attention spatial encoder and an LSTM-based temporal encoder. The spatial encoder treats each NEM region as a node in a graph and learns attention weights for power flows over regions, while the temporal encoder captures autocorrelation within regions. We will then compare the designed model against a set of standard deep learning baselines (MLP, CNN, LSTM, CNN-LSTM, and a self-attention/Transformer model) trained on the same data splits, forecast horizons (day-ahead and two-day-ahead), and evaluation metrics (MAE, RMSE, sMAPE, and directional accuracy), with particular attention to performance during extreme and negative price events. This methodology is inspired by the benchmarking approach of Gani et al. (2026), and directly tests whether an increase in model complexity correlates with better performance on volatile, regime-switching electricity price series, while relating our findings to concepts covered in the course.

2 Significance

Electricity is sold at unregulated spot prices but bought by consumers at regulated rates, so when prices swing wildly, retailers take the hit directly, which can threaten both their finances and the stability of the grid (Li et al., 2025). Between October 2021 and January 2022, average NEM spot prices were almost four times higher than the year before and peaked at $14,000/MWh (Li et al., 2025). More recent data from 2023–2025 shows this hasn't gone away: growing solar uptake now pushes midday prices negative on a large share of intervals, and the evening ramp as solar output drops off is still a major source of extreme price spikes (Gani et al., 2026). Being able to forecast prices accurately in these conditions is useful for retailers managing risk, generators deciding when to bid and system operators trying to keep enough reserve capacity on hand. Gani et al. (2026) found that several recent state-of-the-art time-series models (TimesNet, iTransformer, TimeMixer), despite being top performers on standard, fairly stable benchmark datasets, are often beaten by much simpler recurrent and convolutional models once tested on real NEM data with actual volatility, structural breaks and negative prices. This shows that a bigger or more complex model isn't automatically better; what matters instead is whether a model's assumptions, like explicit spatial structure, sensible loss design, or multi-task regularisation, fit the statistical behaviour of the data it's being used on.

3 Feasibility

Data. Both reference papers draw on data from AEMO, which publishes historical Regional Reference Price (RRP), demand, generation, and inter-regional power-flow data at 5-minute resolution for all five NEM regions. Gani et al. (2026) retrieve this data programmatically and share their processed dataset and code on GitHub, while Li et al. (2025) use an equivalent feature set. We will use `NEMSEER`, a Python package for querying and pre-processing AEMO NEM data, to pull and clean these same series for all five regions and build a matching feature set, giving us a large, recent, license-clear dataset without writing custom scraping code.

What has already been done. Over the past four weeks we have reviewed both papers in detail, including their data pipelines, architectures, and evaluation methodology, confirmed that the code and data of Gani et al. (2026) are openly available as a reproducibility reference, and set up a `NEMSEER`-based data pipeline as the first step of our own implementation.

Planned work. We have split into parallel workstreams and organised the implementation into two-week sprints, so each stage exercises concepts covered in lectures. Sprint 1 (Weeks 6–7, 31/08–13/09) covers data retrieval and preprocessing with `NEMSEER`; implementation of an MLP, a Temporal Convolutional Network, a Transformer, an LSTM, and a Gradient Boosting baseline as a non-deep-learning reference point; a shared evaluation pipeline reporting MAE, RMSE, sMAPE, and directional accuracy so every model is scored consistently; and finalising this proposal. This spans course content from data preparation and model selection through to gradient-based learning, convolutional and recurrent architectures, and attention mechanisms. Sprints 2–3 (Weeks 8–9) then consolidate evaluation across the Sprint 1 models and move on to the harder graph-attention multi-task model inspired by Li et al. (2025), applying course content on optimisation and regularisation before building the graph-neural-network layers and the multi-task loss with learned task weights.

Evaluation. We will report MAE, RMSE, sMAPE, and mean directional accuracy on day-ahead and two-day-ahead horizons, with a dedicated breakdown for the upper/lower 5% price tail and negative-price intervals, following Gani et al. (2026), and weighted-F1 for the auxiliary spike-classification task once the multi-task model is implemented, following Li et al. (2025).

Risks and mitigation. The main risks are compute budget for training six model families across five regions and two horizons, mitigated by training on UQ's Rangpur HPC cluster for GPU resources well beyond a personal machine, and the added complexity of the graph-attention model, which is deliberately scheduled for Weeks 8–9 so a working set of simpler baselines and a shared evaluation pipeline already exist first. Given the public data, the `NEMSEER` pipeline, available reference implementations, and the sprint plan's alignment with weekly course content, we consider the project feasible within the semester timeline.

References

Gani, M. O. et al. (2026). "Deep Time-Series Models Meet Volatility: Multi-Horizon Electricity Price Forecasting in the Australian National Electricity Market". In: *arXiv preprint arXiv:2602.01157*.

Li, Y. et al. (Jan. 2025). "Multi-Task Graph Adaptive Learning for Multivariate Electricity Price Short-Term Forecasting in Australia's National Electricity Market". In: *IEEE Transactions on Power Systems* 40.1, pp. 530–541. DOI: 10.1109/TPWRS.2024.3380834.